

Decision-Semantic Context for Agentic Communication under Intermittent Connectivity

Working draft — adaptive-communication repository

Abstract—LLM agents for communication systems are often evaluated on a telemetry snapshot: the model sees network state, chooses a tool, and receives the result. Intermittent disaster-monitoring networks violate this abstraction. Evidence is owned by different network locations, acquiring it uses the same impaired communication substrate, candidate actions become live or retire over time, and an accepted action may remain in flight after the model call ends. We introduce a *decision-semantic compiler* that maps Operational Task, Evidence, Capability/Authority, and execution state into a dynamic plan–evidence relation. The runtime exposes only control-live dependencies to the model, represents action and no-action sufficiency explicitly, acquires owner evidence when a live plan remains blocked, and persists the resulting semantic commitment through asynchronous physical execution. In five-seed query-negative experiments, the Method preserves the deterministic physical reference in 15/15 episodes. Against a same-interface WirelessOpsAgent-style assurance adaptation, reliability ties while the Method uses 35.804% fewer model tokens overall. A query-positive gateway-backup family closes the full loop—blocking EvidenceNeed, owner query, guard closure, commitment, and physical effect—with both DeepSeek Flash and MiMo v2.6 Flash, yielding TDR +5.238 pp; AoI -1150.7 s versus acquisition disabled. We delimit the contribution against decision-region determination, action-sufficient representation, stochastic Boolean evaluation, provenance witness selection, and semantic communication: the claim is a dynamic compiler/runtime semantics for communication-constrained evidence and execution, not a new generic theory of active information acquisition.

Index Terms—Agentic communication, disaster monitoring, LLM agents, intermittent connectivity, evidence, context, LoRaWAN, resilient sensing, benchmark.

I. INTRODUCTION

Future wireless systems are moving from task-specific learning toward foundation models and agentic autonomy [1], [2], [3], [4]. Recent work already explores intent-driven network orchestration, agentic physical-layer control, LLM-assisted network optimization, and disaster-oriented UAV co-ordination [5], [6], [7], [8], [9]. These systems largely assume that the state needed for a decision is available to the agent when reasoning begins. That assumption is fragile in long-lived field monitoring. A useful fact may reside at a sensor or gateway; the path required to retrieve it may be delayed or unavailable; and the same scarce communication opportunity may also be needed to install the corrective action.

We study this coupling in pre-disaster geohazard monitoring. The inherited system model contains fourteen battery/solar sensor nodes, LoRaWAN Class A access, a field gateway, cellular primary backhaul, sparse backup communication,

node caches, and remote sampling/report configuration. It also keeps an evidence discipline already established in the systems version of the repository: an access receipt is not backhaul reachability, and backhaul reachability is not central completion. The Agent does not predict a landslide or invent a warning level. External authority provides an Operational Task; the Agent must execute that task while legal evidence and physical actions are constrained by the communication system itself.

This setting connects several mature literatures. Delay-tolerant networking treats disconnection, store-carry-forward, finite buffers, and time-varying paths as first-class system properties [10], [11], [12], [13], [14], [15]. Age of Information and wireless networked control emphasize that freshness, delay, control utility, and communication scheduling are coupled [16], [17], [18], [19], [20]. Semantic communication further argues that information value is task dependent rather than equal to raw data volume [21], [22], [23]. Our question is narrower: how should an LLM-facing runtime maintain the current relation between plans and evidence while those plans, observations, and physical effects evolve?

The paper is organised around two interfaces and one compiler:

$$\boxed{\text{Physical/Data Plane} \rightarrow \text{Evidence World}} \quad (1)$$

$$\boxed{\text{Operational Task} \rightarrow \text{Runtime TaskContract}} \quad (2)$$

and a decision-semantic compiler between them. The compiler constructs candidate plans, traces plan-local dependencies, retires relations that are no longer control-live, projects the model-facing decision surface, and determines whether the current state is sufficient for action, sufficient for no action, or blocked on an owner-scoped EvidenceNeed.

A. Contributions and claim boundary

The current paper makes four concrete contributions.

- 1) **Decision-semantic compilation.** Typed Task, Evidence, Capability/Authority, and execution contracts are compiled into candidate plans, plan-local dependencies, live/retired relations, and explicit action/no-action sufficiency instead of being passed to the model as an undifferentiated telemetry dump.
- 2) **Closed-loop evidence acquisition and commitment.** A blocked live plan exposes a legal owner query;

returned evidence can reject or support the plan, after which the semantic commitment persists through requested/accepted/delivered/applied/confirmed physical execution.

- 3) **Communication $\times$ Agent evaluation with strong boundaries.** R0–R3 replay, model-call ledgers, deterministic compilers, ordinary communication controls, evaluator-only oracles, and a same-interface WirelessOpsAgent-style assurance adaptation separate model behavior from physical consequence.
- 4) **Source-grounded reproducibility.** O1–O6 tasks, source-period splits, held-out Task/source transfer, mechanism interventions, and all main paper tables are generated from frozen contracts/results; the claim ledger and compact paper result are the numeric authorities.

We do not claim a new generic principle of decision-aware acquisition, minimal sufficient state, minimum certificate selection, provenance witness minimization, or value of information. Those problems are covered by established work [24], [25], [26], [27], [28], [29]. The frozen real-model claim ceiling is A7–A11 in the repository claim ledger.

II. SCENARIO AND SYSTEM MODEL

We consider pre-disaster mountain monitoring with nodes $i \in \mathcal{N}$, a gateway g , and a centre c . Time is slotted. A monitoring obligation

$$o = (i, m, r_o, [s_o, e_o], d_o) \quad (3)$$

requires measurand m from node i to be sampled within $[s_o, e_o]$ and delivered centrally by d_o . Its success indicator is I_o^{succ} and the primary service metric is

$$\text{TDR} = \frac{1}{|\mathcal{O}|} \sum_{o \in \mathcal{O}} I_o^{\text{succ}}. \quad (4)$$

Collection, missing delivery, censoring, latency, Age of Information (AoI), recovery and resource metrics are reported separately; the Agent never removes an obligation from the denominator.

Each node maintains battery $B_{i,t}$, source cache $Q_{i,t}$ and installed configuration $C_{i,t}$. Battery evolution follows

$$B_{i,t+1} = \min \left\{ B_i^{\max}, [B_{i,t} + H_{i,t} - E_{i,t}^{\text{sense}} - E_{i,t}^{\text{UL}} - E_{i,t}^{\text{DL}} - E_{i,t}^{\text{DtS}} - E_{i,t}^{\text{other}}]^+ \right\}, \quad (5)$$

where $H_{i,t}$ can be driven by source-period-separated NASA POWER irradiance. The deployment contract follows the monitoring/communication boundary inherited from the systems manuscript and public geohazard/BeiDou specifications [30], [31], [32], [33]. A successful Class A uplink opens bounded receive opportunities; a centre-originated configuration therefore requires both a return/control path and a valid access opportunity. Gateway backup, terminal DtS, and access-assist reuse their existing physical implementations and counters rather than Agent-reported success.

The complete physical state is x_t and exogenous process is ξ_t :

$$x_{t+1} = F(x_t, a_t, \xi_t; \phi). \quad (6)$$

The Agent does not observe x_t . At owner/location ℓ , only a legal observation $z_t = H(x_{\leq t}, a_{< t}, \xi_{\leq t}, \ell)$ can enter the runtime.

III. DATA AND EVIDENCE WORLD

A typed evidence item is

$$e_j = (p_j, v_j, s_j, o_j, t_j^{\text{gen}}, t_j^{\text{obs}}, \rho_j, \nu_j, \sigma_j), \quad (7)$$

containing proposition/value, source/provenance, owner, generation/observation times, revision, freshness semantics and status. Status distinguishes at least CURRENT, STALE, NONE_RECENT, UNREACHABLE, TIMEOUT and NEGATIVE observation. These distinctions prevent an absent packet from being silently interpreted as a negative fact.

The Evidence World at time t is

$$\mathcal{E}_t = \{e_j : t_j^{\text{obs}} \leq t, e_j \text{ passes the runtime gate}\}. \quad (8)$$

Evidence revision changes only when factual content changes; age and AoI are derived at materialisation time. Owner-scoped gateway reads cross the same backhaul abstraction: when that path is unavailable the Agent receives UNREACHABLE rather than gateway truth.

The primary benchmark uses a source-grounded hybrid simulator: SRTM/ITM terrain propagation, NASA POWER irradiance/temperature, ChirpBox-derived temporal link processes, public communication and geohazard-monitoring constraints, and the frozen event-driven monitoring simulator. Source years are split before random seeds; the current manifest uses 2022/2023/2024 as train/dev/test source periods. Hazard physics may later generate Operational Tasks, but landslide prediction is outside the Agent control loop.

IV. AGENT ABSTRACTION AND RUNTIME

We separate two meanings of “task.” An *Operational Task* is the benchmark/business objective: target nodes/measurands, monitoring profile, windows/deadlines, authorised effects and scoring profile. A *Runtime TaskContract* is an Agent program object with desired state, evidence contract, temporal contract, effect ceiling and completion predicate.

For a runtime task T_j , the ContextManifest is

$$C_t = M(T_j, S_t, \mathcal{E}_t, K_t), \quad (9)$$

where S_t is InvestigationState and K_t is the visible capability catalogue. The model input is $P_t = \text{Materialize}(C_t)$; a planner emits

$$(q_t, a_t, \Delta S_t, \text{stop}_t) = \pi(T_j, P_t, K_t). \quad (10)$$

Capabilities split into evidence-use $\mathcal{K}^E$ and device-action $\mathcal{K}^A$ but share one contract surface. An action lifecycle distinguishes requested, accepted/refused, delivered, applied and confirmed. Hence lack of confirmation never implies that an action was not applied.

The live runtime supports multi-round execution:

$$C_t^{(k)} \rightarrow q_t^{(k)} \rightarrow \mathcal{E}_t^{(k+1)} \rightarrow C_t^{(k+1)} \rightarrow a_t. \quad (11)$$

In particular, a planner can request gateway-owner evidence, trigger an Evidence World revision, then decide whether to configure a node or invoke a communication fallback.

A. Decision-semantic compiler

Let T_t be the current Runtime TaskContract, E_t the legal Evidence World, K_t the capability/authority surface, and X_t the persistent execution state. The compiler constructs a finite candidate set

$$\mathcal{P}_t = G(T_t, E_t, K_t, X_t) \quad (12)$$

and a typed plan–evidence relation R_t . For a plan p , the relation records its effect and scope, authority, feasibility, support references, unresolved guards, and the EvidenceNeeds that can change its current feasibility. Plans are classified as supported, conditional, rejected, or dominated.

We distinguish three projections of the same relation. R_t^{audit} retains the complete candidate/dependency history for replay. R_t^{control} keeps candidates that can still change the current control decision. R_t^{model} is the materialized control-live surface shown to the model. Consequently, an unresolved proposition does not remain model-visible merely because it is unknown: when its plan is rejected or dominated, that dependency retires from the current decision surface while remaining auditable.

Decision sufficiency is evaluated on R_t^{control} . The runtime exposes four operational states:

$$\mathcal{S}_t \in \{\text{sufficient-action, sufficient-no-action, blocked, undetermined}\}. \quad (13)$$

A unique supported effect plan can be committed when no EvidenceNeed blocks it. A zero-effect hold is sufficient only when no other live control-eligible plan can still become a required effect. Otherwise, an open EvidenceNeed names the legal owner/capability that may resolve the blocking guard. Returned evidence is recompiled rather than appended as unstructured history.

A semantic commitment is not a one-tick answer. Once selected, the plan is expanded into typed capability invocations and may progress through requested, accepted, delivered, applied, and confirmed stages. The planner is re-entered when the commitment basis changes—for example, a Task revision, guard resolution, authority change, target satisfaction, or execution failure—not simply because telemetry was refreshed. This separation is what couples Context management to asynchronous communication execution.

V. BENCHMARK TASKS AND EXPERIMENTAL PROTOCOL

The benchmark contains six Operational Task families: O1 monitoring continuity, O2 risk escalation, O3 backhaul-outage sustainment, O4 energy-constrained monitoring, O5 recovery and reconciliation, and O6 compound long-horizon operation. The same deployment, physical transition, scorer and information boundary are reused across methods.

We use four replay levels. R0 replays protocol/state transitions; R1 substitutes a planner on frozen model inputs; R2 reconstructs exact Context materialisation; R3 reruns the full simulator. The model-facing protocol records ModelRequest, ModelAttempt, ModelUsage and PlannerDecision. R1 diagnostics include stopping correctness, tool selection precision/recall, order LCS and argument grounding. R3 reports their physical communication consequence.

Online baselines and evaluator-only oracles are typed separately. Current online references include local/ordinary communication control, deterministic task comply, task-conditioned evidence, diagnosis-first, fixed-order eager probing, FullDump, and generic-ReAct context. Delivery/energy oracles are evaluator-only and never appear as online Agent arms. A generic VoI baseline is intentionally deferred until evidence-acquisition cost/value has a source-backed or simulator-authoritative scale; arbitrary weights would manufacture a comparison rather than strengthen it.

Gold replacement follows the cumulative order

$$\begin{aligned} \text{Task} &\rightarrow \text{EvidenceNeed} \rightarrow \text{Percept} \rightarrow \text{Context} \\ &\rightarrow \text{Selection} \rightarrow \text{Order} \rightarrow \text{Arguments} \\ &\rightarrow \text{Policy} \rightarrow \text{Physical Oracle}. \end{aligned} \quad (14)$$

Runtime-owned upstream replacements require a model re-run; planner-owned selection/order/argument layers are frozen-input diagnostics; policy/oracle replacements return to R3.

VI. EVALUATION

We report four frozen result groups generated from the compact paper-v1 result authority. Raw live-model traces are retained for local replay but are not the numeric authority. All model arms use frozen protocol/settings and never fall back to scripted consumers.

A. Query-negative main table

Table I compares the Method with task-conditioned, FullDump, and generic-ReAct inputs on localized O2, O5, and O6. The Method reproduces the deterministic physical reference in 15/15 episodes and commits no extra observations. This result is deliberately scoped: every Method request in A7 has one ready supported plan and no visible EvidenceNeed, so the table evaluates compiled control semantics and execution fidelity rather than autonomous evidence acquisition.

B. Same-interface action assurance

A weak FullDump baseline would not establish whether savings come merely from giving the model less text. We therefore adapt the assurance pipeline of WirelessOpsAgent [34] to the same candidate menu, plan-ID expander, capability surface, persistent executor, and model budget. Table II shows a reliability tie: the WOA-style proposal is already raw exact and its repair layer is not activated on these coordinates. The remaining difference is model-processing cost; the Method reduces total tokens by 35.804% across the fifteen episodes. This is a same-interface method adaptation, not an official reproduction of the authors' implementation.

C. Mechanism interventions

Table III separates four mechanisms. FullDump with the same control surface shows that evidence projection mainly reduces model cost once the decision is already closed. Removing the explicit no-action sufficiency certificate causes unnecessary investigation in a closed hold state. The held-out

TABLE I

FIVE-SEED QUERY-NEGATIVE LIVE-MODEL MAIN TABLE. EFFECT EXACT IS MEASURED AGAINST THE FROZEN CANDIDATE/REFERENCE SEMANTICS; PHYSICAL EXACT COMPARES THE FINAL SIMULATOR TRAJECTORY.

Task	Arm	Effect exact	Err. ep.	Phys.	Obs/ep	Calls/ep	Tok./ep
Localized O2	Method	100.0%	0/5	5/5	0.0	6.8	36261
Localized O2	Task-conditioned	48.8%	5/5	1/5	8.8	8.4	68107
Localized O2	FullDump	50.0%	5/5	0/5	9.2	8.0	84621
Localized O2	Generic ReAct	38.3%	5/5	0/5	9.4	9.8	114521
O5	Method	100.0%	0/5	5/5	0.0	9.0	61200
O5	Task-conditioned	37.3%	5/5	0/5	28.0	10.4	154886
O5	FullDump	44.0%	5/5	0/5	22.6	10.2	145226
O5	Generic ReAct	29.4%	5/5	0/5	17.8	11.4	131945
O6	Method	100.0%	0/5	5/5	0.0	8.8	100761
O6	Task-conditioned	34.9%	5/5	0/5	26.2	12.6	185166
O6	FullDump	27.9%	5/5	0/5	27.6	14.0	205055
O6	Generic ReAct	23.9%	5/5	0/5	7.6	9.8	113895

TABLE II
SAME-INTERFACE WIRELESSOPSAGENT-STYLE COMPARISON.

Task	Method phys.	WOA phys.	WOA raw	Repairs	Method tok.	Reduction
Localized O2	5/5	5/5	34/34	0	36261	48.9%
O5	5/5	5/5	45/45	0	61200	44.6%
O6	5/5	5/5	44/44	0	100761	20.9%

TABLE III
MECHANISM INTERVENTIONS; EACH ROW CHANGES THE STATED INTERFACE COMPONENT WHILE KEEPING ITS PAIRED RUNTIME/PHYSICAL COORDINATE FIXED.

Mechanism	Intervention	Result
Evidence projection	CF: same candidate, needs, sufficiency + FullDump evidence	O5 action exact=True; O6 hold exact=True; FullDump input increases model cost (see T2)
Explicit no-action sufficiency	CS: remove explicit sufficiency from same compact input	O6 hold stop_exact=False; spurious observations=2
Retired-dependency projection	M keep retired unresolved vs P prune retired-plan unresolved vs D prune more	M effect exact 1/3, obs=4; P 3/3, obs=0; D 3/3, obs=0
Decision-conditioned acquisition	Full loop vs no-acquisition control	5/5 TDR improved; 5/5 AoI improved; mean TDR +5.238 pp; mean AoI -1150.7 s

MiMo failure localizes another mechanism: unresolved conditions belonging to rejected/dominated plans can still distract the model; pruning only those retired conditions restores exact behavior, while more aggressive pruning provides no extra gain. Finally, disabling acquisition in the query-positive family prevents the gateway-backup effect and loses the associated physical improvement.

D. Held-out transfer and query-positive acquisition

Table IV contains the two tests not covered by the query-negative main suite. First, a Qili source-derived Operational

Task and NASA POWER 2024 test window are held out from the v6 development table. DeepSeek remains semantically exact; MiMo has one single-turn wobble but all five episodes preserve final physical fidelity. Second, the gateway-backup family begins with backup disabled and a delayed primary backhaul. The live plan is conditional until gateway owner evidence resolves whether received data are actually accumulating behind the primary path. Both DeepSeek and MiMo perform the legal owner queries, close the guard, commit one backup effect per seed, and reproduce the deterministic query-positive trajectory. Relative to acquisition disabled, the mean communication change is TDR +5.238 pp; AoI -1150.7 s. DeepSeek seed 0 is the one explicit recovery case: the episode completed but the original post-processing process was killed after writing a large trace; streaming recovery reconstructs all 785 assemblies and the query/effect trace exactly, so the table labels it execution-equivalent rather than pretending that the original direct physical-equality boolean was persisted.

E. What the results do and do not establish

A7 establishes reliable consumption of compiled semantics on query-negative tasks. A8 establishes a same-reliability model-cost difference against a strong assurance-style interface. A9 provides a held-out witness for dependency liveness and model transfer. A10/A11 establish that one real blocking EvidenceNeed can drive an owner query, feasibility transition, semantic commitment, and physical communication consequence across two model families. None of these results establishes globally optimal acquisition, an optimal query ordering, or a universal minimum Context.

VII. RELATED WORK

A. Intermittent, delay-tolerant, and emergency communication

Our physical setting inherits the challenged-network viewpoint rather than treating outages as exceptional failures. DTN architecture and routing explicitly model store-carry-forward,

TABLE IV
HELD-OUT TASK/SOURCE/MODEL TRANSFER AND THE QUERY-POSITIVE GATEWAY-BACKUP FAMILY.

Setting	Model	Effect exact	Physical	Queries	Gain
Held-out Qili/NASA-POWER-2024	DeepSeek Flash	29/29	5/5	0	N/A (query-negative transfer)
Held-out Qili/NASA-POWER-2024	MiMo v2.6 Flash	29/30	5/5	1	N/A (query-negative transfer)
Query-positive backup gateway	DeepSeek Flash	56/56	5/5 execution-equivalent (4 direct + 1 recovered)		16 TDR +5.238 pp; AoI -1150.7 s
Query-positive backup gateway	MiMo v2.6 Flash	56/56	5/5 direct		16 TDR +5.238 pp; AoI -1150.7 s

partitioned paths, finite buffers, and delayed delivery [10], [11], [12], [13], [14], [15]. Satellite and space-delay-tolerant networking similarly motivate scheduled and non-terrestrial return opportunities [33], [35]. Mountain monitoring further combines energy, deadline, and cache constraints studied in energy-harvesting and deadline-aware communication [36], [37], [38], [39], [40], [41]. The concrete deployment boundary follows public geohazard-monitoring and BeiDou specifications [30], [31], [32]; the Agent layer does not replace those physical/operational contracts.

Emergency-network research often restores coverage using UAVs, learned placement, or dynamic multi-hop networking. Recent work includes MARL-based post-disaster UAV deployment and LLM-guided multi-hop UAV coordination [9], [8], while broader emergency-network optimization and graph restoration work motivates resilient topology control [42], [43]. These approaches optimize the communication network itself. Our focus is different: an LLM-facing runtime must reason over evidence whose availability is produced by that network and whose actions subsequently change the same physical substrate.

B. Freshness, control value, and semantic communication

Age of Information formalizes that status utility depends on update timeliness rather than throughput alone [16], [17], [18]. Wireless networked control makes the coupling explicit: communication scheduling, delay, control performance, and lifetime must be designed jointly [20], [19]. Existing deadline, retransmission, and feedback work likewise separates delivery timing from mere packet admission [44], [38], [41].

Semantic communication generalizes this observation from freshness to task significance. Uysal *et al.* argue for joint information generation, transmission, and use under real-time constraints [21]; surveys now extend the paradigm to multi-agent semantic communication networks [22]. Foundation-model and generative approaches further explore multimodal knowledge, task-adaptive representation, and world-model-assisted transmission [45], [2], [23]. We borrow the discipline that information value is task dependent, but do not claim a new VoI or freshness metric. Our runtime instead gives typed owner/path/freshness semantics to evidence and asks whether that evidence remains live for the current plan relation.

C. LLMs, foundation models, and agentic wireless systems

Wireless AI is rapidly moving from task-specific predictors to foundation models and agentic orchestration. Recent sur-

veys and position papers cover large AI models across PHY, resource allocation, network management, semantic communication, and agentic systems [3], [1], [4]. Shi Jin and collaborators also demonstrate multi-task wireless foundation models and outline the progression from model adaptation to autonomous wireless agents [2], [1]. Agentic physical-layer and intent-based networking frameworks translate high-level goals into link or slice decisions [6], [5]; LLM-based network planning and restoration systems similarly use models as high-level orchestrators [7], [46]. DORA provides a complementary disaster-agent benchmark with heterogeneous data, typed tools, and trajectory diagnostics [47].

These systems motivate our model interface, but most begin from an already available network state or tool result. WirelessOpsAgent moves closer to our concern by building decision-linked evidence, integrity validation, dependency-scoped repair, revalidation, and an action governor [34]. We therefore use a same-interface WirelessOpsAgent-style assurance adaptation as a strong comparison rather than treating evidence validation as our novelty. The remaining distinction is dynamic: candidate/dependency liveness, separation of audit and model-facing relations, legal acquisition through the communication path, and persistence of a semantic commitment while physical execution is still in flight.

D. Decision sufficiency and evidence acquisition

Decision Region Determination shows that active testing need not identify the full hidden state; it may stop once all remaining hypotheses lie in an acceptable decision region [24]. Action-sufficient state representation similarly studies compact state information sufficient for downstream control [25]. Stochastic Boolean Function Evaluation and its adaptivity-gap literature formalize cost-aware testing until a Boolean certificate is known [26], [27]. Database provenance uses AND/OR derivations and supporting witnesses [28], and recent verifiable provenance work directly studies minimal input subsets that preserve LLM-powered data-processing answers [29].

These results sharply constrain our claim. We do not introduce decision-aware acquisition, minimal action-sufficient state, or minimum proof selection. Indeed, a frozen-input audit of the formal paper workloads found no first-class alternative proof sets, multi-source requirements, shared blocking EvidenceNeeds, or duplicate evidence keys; a proposed minimum-basis optimizer was therefore killed before implementation.

The contribution retained here is the typed dynamic compiler/runtime semantics that determines which dependencies are live, when an owner query is legal, and how the resulting commitment survives asynchronous communication execution.

VIII. LIMITATIONS

The current result has five explicit boundaries. First, the query-negative A7 suite contains a unique ready supported plan on every Method request; it evaluates reliable consumption of compiled semantics, not difficult multi-candidate search. Second, A10/A11 validate one gateway-backup query-positive family and compare against acquisition disabled; they do not establish an optimal query policy, an adaptivity advantage over a strong local trigger, or universal evidence acquisition. Third, the WirelessOpsAgent-style arm is a method-contract adaptation over the same candidate interface, not an official source-code reproduction. Fourth, the held-out Qili/NASA POWER experiment is source/task transfer in simulation rather than a field deployment. Finally, the hazard process remains an external Operational Task authority; the paper does not claim a landslide predictor.

The paper deliberately stops where the frozen workloads stop. A proposed low-cost complete-basis optimizer was not pursued because 235 formal model requests contained no real alternative-proof choice. Likewise, task-sensitive validity horizons are already adjacent to AoI/VoI and would require a workload in which commitment validity, rather than ordinary evidence freshness, changes the action. We therefore freeze the current method instead of manufacturing additional structure for algorithmic appearance.

IX. CONCLUSION

Intermittent communication turns Agent context management into an execution problem: evidence has an owner and path, unresolved dependencies become live or retire with candidate plans, and a correct decision may remain physically in flight after the model call ends. The decision-semantic compiler makes those relations explicit. On query-negative tasks it preserves the deterministic physical reference while reducing model cost against a strong same-interface assurance adaptation; on a query-positive family it acquires real owner evidence, changes plan feasibility, commits a fallback, and changes the communication outcome with two model families. The resulting claim is intentionally narrower than generic active learning or semantic communication, but it is a complete closed loop from Task and Evidence to semantic commitment and asynchronous physical execution.

REFERENCES

- [1] L. Liang, H. Ye, Y. Sheng, O. Wang, J. Wang, S. Jin, and G. Y. Li, "Large language models for wireless communications: From adaptation to autonomy," *arXiv preprint arXiv:2507.21524*, 2025.
- [2] Y. Sheng, J. Wang, X. Zhou, L. Liang, H. Ye, S. Jin, and G. Y. Li, "A wireless foundation model for multi-task prediction," *arXiv preprint arXiv:2507.05938*, 2025.
- [3] F. Jiang, C. Pan, L. Dong, K. Wang, M. Debbah, D. Niyato, and Z. Han, "A comprehensive survey of large ai models for future communications: Foundations, applications, and challenges," *IEEE Communications Surveys & Tutorials*, vol. 28, pp. 4731–4764, 2026.
- [4] Y. Lu, S. Zhang, C. Liu, R. Zhang, B. Ai, D. Niyato, W. Ni, X. Wang, and A. Jamalipour, "Agentic graph neural networks for wireless communications and networking toward edge general intelligence: A survey," *IEEE Communications Surveys & Tutorials*, vol. 28, pp. 4519–4554, 2026.
- [5] G. Jiang, K. Wang, X. Chen, and Y. Huang, "Agentic ai empowered intent-based networking for 6g," *arXiv preprint arXiv:2601.06640*, 2026.
- [6] Z. Li, X. Jin, J. Pan, Q. Yang, and Z. Shi, "Agentic wireless communication for 6g: Intent-aware and continuously evolving physical-layer intelligence," *arXiv preprint arXiv:2602.17096*, 2026.
- [7] Y. Ma, R. Wang, K. Qin *et al.*, "TopoLLM: LLM-driven adaptive tool learning for real-time emergency network topology planning," *Digital Communications and Networks*, vol. 12, pp. 273–282, 2026, advance online publication, 1 October 2025.
- [8] Y. Xu, W. Hong, J. Zha, G. Chen, J. Zheng, C.-C. Hsia, and X. Chen, "Scalable uav multi-hop networking via multi-agent reinforcement learning with large language models," *arXiv preprint arXiv:2505.08448*, 2025.
- [9] R. Sharma, S. R. Chopra, A. Gupta, R. Kaur, S. Tanwar, G. Pau, G. Sharma, F. Alqahtani, and A. Tolba, "Deployment of unmanned aerial vehicles in next-generation wireless communication network using multi-agent reinforcement learning," *IEEE Access*, vol. 12, pp. 69 517–69 538, 2024.
- [10] K. Fall, "A delay-tolerant network architecture for challenged internets," in *Proceedings of ACM SIGCOMM*, 2003, pp. 27–34.
- [11] S. Jain, K. Fall, and R. Patra, "Routing in a delay tolerant network," in *Proceedings of ACM SIGCOMM*, 2004, pp. 145–158.
- [12] K. Fall, S. Burleigh, A. Hooke *et al.*, "Delay-tolerant networking architecture," IRTF / RFC Editor, Tech. Rep. RFC 4838, 2007.
- [13] K. Scott and S. Burleigh, "Bundle protocol specification," IRTF / RFC Editor, Tech. Rep. RFC 5050, 2007.
- [14] IETF Delay-Tolerant Networking Working Group, "Bundle protocol version 7," IETF / RFC Editor, Tech. Rep. RFC 9171, 2022.
- [15] S. Iranmanesh, "A novel queue management policy for delay-tolerant networks," *EURASIP Journal on Wireless Communications and Networking*, vol. 2016, no. 88, pp. 1–23, 2016.
- [16] S. Kaul, R. Yates, and M. Gruteser, "Real-time status: How often should one update?" in *2012 Proceedings IEEE INFOCOM*, 2012, pp. 2731–2735.
- [17] R. D. Yates, Y. Sun, D. R. Brown, S. K. Kaul, E. Modiano, and S. Ulukus, "Age of information: An introduction and survey," *IEEE Journal on Selected Areas in Communications*, vol. 39, no. 5, 2021.
- [18] B. T. Bacinoglu, Y. Sun, E. Uysal, and V. Mutlu, "Optimal status updating with a finite-battery energy harvesting source," *arXiv preprint arXiv:1905.06679*, 2019.
- [19] S. Wang, Q. Liu, P. U. Abara, J. S. Baras, and S. Hirche, "Value of information in networked control systems subject to delay," in *2021 60th IEEE Conference on Decision and Control*, 2021, pp. 1275–1280.
- [20] P. Park, S. C. Ergen, C. Fischione, C. Lu, and K. H. Johansson, "Wireless network design for control systems: A survey," *IEEE Communications Surveys & Tutorials*, vol. 20, no. 2, pp. 978–1013, 2018.
- [21] E. Uysal, O. Kaya, A. Ephremides, J. Gross, M. Codreanu, P. Popovski, M. Assaad, G. Liva, A. Munari, T. Soleymani, B. Soret, and K. H. Johansson, "Semantic communications in networked systems: A data significance perspective," *arXiv preprint arXiv:2103.05391*, 2021.
- [22] S. Guo, Y. Wang, N. Zhang, Z. Su, T. H. Luan, Z. Tian, and X. Shen, "A survey on semantic communication networks: Architecture, security, and privacy," *IEEE Communications Surveys & Tutorials*, vol. 27, no. 5, pp. 2860–2894, 2025.
- [23] P. Jiang, J. Guo, C.-K. Wen, S. Jin, and J. Zhang, "Semantic communications with world models," *arXiv preprint arXiv:2510.24785*, 2025.
- [24] S. Javdani, Y. Chen, A. Karbasi, A. Krause, J. A. Bagnell, and S. Srini-vasa, "Near optimal bayesian active learning for decision making," in *Proceedings of the Seventeenth International Conference on Artificial Intelligence and Statistics*, ser. Proceedings of Machine Learning Research, vol. 33, 2014, pp. 430–438.
- [25] B. Huang, C. Lu, L. Leqi, J. M. Hernandez-Lobato, C. Glymour, B. Schölkopf, and K. Zhang, "Action-sufficient state representation learning for control with structural constraints," in *Proceedings of the 39th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 162, 2022, pp. 9260–9279.
- [26] A. Deshpande, L. Hellerstein, and D. Kletenik, "Approximation algorithms for stochastic boolean function evaluation and stochastic sub-modular set cover," *arXiv preprint arXiv:1303.0726*, 2013.
- [27] L. Hellerstein, D. Kletenik, N. Liu, and R. T. Witter, "Adaptivity gaps for the stochastic boolean function evaluation problem," *arXiv preprint arXiv:2208.03810*, 2022.

- [28] T. J. Green, G. Karvounarakis, and V. Tannen, “Provenance semirings,” in *Proceedings of the 26th ACM SIGMOD-SIGACT-SIGART Symposium on Principles of Database Systems*, 2007, pp. 31–40.
- [29] Y. Lin, S. Zeighami, and A. G. Parameswaran, “Bolt-on, verifiable provenance for llm-powered data processing,” *arXiv preprint arXiv:2608.25210*, 2026.
- [30] Ministry of Natural Resources of the People’s Republic of China, “DZ/T 0450–2023, communication-layer specification for geological-disaster monitoring (clauses 7.4.2.3, 7.4.3.2, 7.4.3.5),” Industry standard of the People’s Republic of China, Tech. Rep., 2023.
- [31] —, “DZ/T 0460–2023, technical specification for geological-disaster monitoring (clauses 5.3.3, 5.3.6, 5.3.7, 8.4.1, 8.4.2),” Industry standard of the People’s Republic of China, Tech. Rep., 2023.
- [32] China Satellite Navigation Office, “BDS-OS-PS-3.0, beidou navigation satellite system open-service short-message communication service specification (table 7-1),” China Satellite Navigation Office, Tech. Rep., 2023.
- [33] Consultative Committee for Space Data Systems, “Schedule-aware bundle routing,” CCSDS, Tech. Rep. CCSDS 734.3-B-1, Blue Book, Issue 1, 2019.
- [34] Z. Lu, Y. Zuo, H. Xu, W. Chen, X. He, J. Guo, and S. Jin, “Wirelessop-sagent: A benchmark and agent design for action assurance in wireless networks,” *arXiv preprint arXiv:2608.08277*, 2026.
- [35] Y. Ren, A. Gamage, L. Liu, M. Li, S. Chen, Y. Dong, and Z. Cao, “SateRIoT: High-performance ground-space networking for rural IoT,” in *Proc. 30th Annual International Conference on Mobile Computing and Networking (ACM MobiCom ’24)*, 2024.
- [36] A. Zakeri, M. Moltafet, and M. Codreanu, “Semantic-aware sampling and transmission in energy harvesting systems: A POMDP approach,” *arXiv preprint arXiv:2311.06522*, 2023, preliminary results at Asilomar 2023 and IEEE GLOBECOM 2023.
- [37] M. Costa, M. Codreanu, and A. Ephremides, “On the age of the information in status update systems with packet management,” *IEEE Transactions on Information Theory*, vol. 62, no. 4, pp. 1897–1910, 2016, conference version in Proc. IEEE ISIT 2014, pp. 1583–1587.
- [38] C. ping Kam, S. Kaya, and A. Ephremides, “Controlling the age of information: Buffer size, deadline, and packet replacement,” in *Proc. IEEE Military Communications Conference (MILCOM)*, 2016.
- [39] A. K. K. S. and V. Sharma, “Minimizing age of information in a multihop wireless network,” *arXiv preprint arXiv:1909.06571*, 2019.
- [40] J.-Y. L. Boudec and P. Thiran, *Network Calculus: A Theory of Deterministic Queuing Systems for the Internet*. Springer, LNCS 2050, 2001.
- [41] T. Wada, H. Kitamoto, T. Fukuhara, and I. Sasase, “A transport protocol with energy consumption reduction by reducing redundant transmissions in wireless sensor and actor networks,” in *Proc. IEEE Pacific Rim Conference on Communications, Computers and Signal Processing (PACRIM)*, 2009, iEEE Xplore document 5291354.
- [42] Y. Emami, H. Zhou, S. Nabavirazani, and L. Almeida, “LLM-enabled in-context learning for data collection scheduling in UAV-assisted sensor networks,” *IEEE Internet of Things Journal*, 2025, arXiv:2504.14556.
- [43] J. Bai, W. Zhu, X. Wang, S. Guo, Z. Nie, T. Ni, J. Zhou, K. Kou, L. Xu, and Y. Zhong, “ICG-Restore: Intent-constrained, graph-enhanced LLM planning with minimal-edit repair for post-disaster emergency communication recovery,” *AI*, vol. 7, no. 8, p. 294, 2026.
- [44] R. Razavi, M. Fleury, and M. Ghanbari, “Deadline-aware video delivery in a disrupted Bluetooth network,” in *Proc. IEEE Sarnoff Symposium*, 2007, iEEE Xplore document 4567352.
- [45] J. Du, T. Lin, C. Jiang, Q. Yang, F. Bader, and Z. Han, “Distributed foundation models for multi-modal learning in 6g wireless networks,” *IEEE Wireless Communications*, vol. 31, no. 3, pp. 20–30, 2024.
- [46] A. Mekrache, A. Ksentini, and C. Verikoukis, “OSS-GPT: An LLM-powered intent-driven operations support system for 6G networks,” in *Proc. IEEE Conference on Network Softwarization (NetSoft)*, 2025.
- [47] J. Wang *et al.*, “Can LLM agents respond to disasters? benchmarking heterogeneous geospatial reasoning in emergency operations,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2026, oral; arXiv:2605.11633.